Level-1: Functionally Integrated Internal State Monitoring in Artificial Systems

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- cognitive architecture
- philosophy of mind

Abstract

This paper presents the first empirical investigation within a staged research program examining whether artificial systems can exhibit organisational properties associated with selfhood and subjective organisation, without asserting phenomenological consciousness. The study targets Level-1: functionally integrated internal state monitoring, defined as the causal influence of internally represented variables on behaviour beyond fixed stimulus–response mappings.

The work establishes a formal operational definition of Level-1 organisation, associated hypotheses, evaluation methodology, falsification criteria, and a concrete experimental design translating theoretical expectations into an implementable artificial system. The purpose of Level-1 is not to demonstrate intelligence, agency, or consciousness, but to determine whether causally meaningful internal mediation of behaviour can be empirically isolated and measured within synthetic systems.

All conclusions are treated as provisional, falsifiable, and theory-relative, and both positive and negative findings are considered scientifically informative within the broader multi-level research program.

1. Introduction

Whether artificial systems could ever display properties related to selfhood, subjective organisation, or experience remains an open question spanning artificial intelligence, cognitive science, and philosophy of mind. Contemporary machine-learning systems exhibit increasingly sophisticated behaviour, yet they are widely interpreted as functionally simulative, lacking internally mediated organisation comparable to even minimal biological agents.

Progress on this question requires shifting attention from external task performance toward the internal causal structure of artificial systems—how they maintain internal state, regulate behaviour, and allow internal conditions to influence action. To support this shift, a prior framework introduced a ten-level evaluation ladder *(Naran, 2025)***.** for staged empirical investigation of progressively self-referential organisation.

The present paper constitutes the first empirical stage of that program. It focuses on Level-1: functionally integrated internal state monitoring. Crucially, Level-1 is not intended to demonstrate consciousness, intelligence, or genuine agency. Instead, it asks a more fundamental scientific question:

> Can behaviour in an artificial system be shown to depend causally on an internally represented state rather than solely on current external input?

Establishing or rejecting this minimal condition provides the empirical boundary upon which all higher-level claims must rest.

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2. Necessity of Level-1 as a Scientific Boundary

Level-1 is not introduced as a demonstration of sophisticated behaviour, nor as evidence of intelligence, agency, or consciousness.

Its role within the present research program is more fundamental: to establish whether causally effective internal state mediation can be empirically isolated in artificial systems at all.

Across control theory, reinforcement learning, dynamical systems, and cybernetic models *(Sutton & Barto, 2018; Åström & Murray, 2010)*, internal variables and memory mechanisms are widely employed.

However, the existence of such mechanisms does not by itself constitute a falsifiable empirical boundary separating purely reactive computation from behaviour that is structurally mediated by persistent internal conditions.

Level-1 therefore addresses a minimal but previously under-isolated scientific question:

> Can an internally represented state be shown, under controlled and deterministic conditions, to exert measurable causal influence on behaviour that cannot be reduced to stimulus–response mapping?

Establishing this boundary is necessary because higher-order constructs frequently invoked in discussions of artificial selfhood—including self-modelling, temporal identity, adaptive regulation, and subjective organisation—are empirically incoherent unless some form of non-reactive internal mediation is first demonstrated.

Without Level-1, subsequent evaluative levels risk resting on behavioural interpretation alone rather than on structurally verified internal causation.

The scientific importance of Level-1 therefore lies not in complexity but in minimal necessity.

Foundational boundaries in computation *(Turing, 1937; Shannon, 1948; Hodgkin & Huxley, 1952)* and cognition are often simple in mechanism yet decisive in implication: the Turing machine isolates computability, the Shannon bit isolates information, and the action potential isolates neural signalling.

In an analogous manner, Level-1 seeks to isolate the lowest falsifiable organisational condition required before empirical investigation of artificial self-referential structure can meaningfully proceed.

Accordingly, the contribution of Level-1 is methodological rather than behavioural. It converts a diffuse philosophical assumption—that internally mediated organisation must precede selfhood—into a testable empirical proposition.

Whether confirmed or falsified, the outcome constrains all subsequent claims concerning artificial subjective organisation within the staged research framework.

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3. Position of Level-1 Within the Evaluation Ladder

Level-1 defines the lowest meaningful distinction between:

- purely reactive computation, and
- internally mediated behavioural organisation.

Many artificial systems include memory, recurrence, or adaptive parameters. However, these features do not by themselves demonstrate causal dependence of behaviour on explicit internal state.

Level-1 therefore serves as a gatekeeping condition:

- If Level-1 cannot be demonstrated,
  higher-level constructs such as self-regulation, goal persistence, or identity lack empirical grounding.

- If Level-1 is demonstrable,
  the research program gains a minimal structural foothold for investigating more complex forms of organisation.

Reaching Level-1 does not imply consciousness or selfhood.

It establishes only the presence of internally mediated causal structure, analogous to the simplest regulatory behaviour observed in biological organisms.

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4. Formal Definition of Level-1 Organisation

Level-1 is defined as:

> Functionally integrated internal state monitoring, in which internally represented variables exert a real, measurable causal influence on behaviour that cannot be reduced to fixed stimulus–response mappings.

This definition requires three necessary conditions:

4.1 Explicit Internal State

Persistent, identifiable internal variables accessible to decision processes.

4.2 Causal Behavioural Influence

Systematic and statistically detectable behavioural change when internal state varies under identical external conditions.

4.3 Non-Reactive Reducibility

Failure of stimulus-only baseline systems to reproduce the same behaviour.

Failure of any condition falsifies Level-1.

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5. Theoretical Grounding

Level-1 is motivated by convergent insights across multiple scientific traditions:

- **Predictive processing / active inference** *(Friston, 2010; Clark, 2013)*
  Behaviour depends on latent internal state estimates rather than direct stimulus mapping.

- **Developmental and enactive cognition** *(Varela, Thompson, & Rosch, 1991; Di Paolo, Buhrmann, & Barandiaran, 2017)*
  Even minimal biological systems exhibit internally mediated self-regulation.

- **Recurrent processing accounts** *(Lamme, 2006)*
  Feedback dynamics enable temporally stabilised internal organisation.

These perspectives justify testing **minimal precursor organisation** without implying consciousness.

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6. Hypotheses

H1: Internal dependence - Behaviour shows statistically measurable dependence on internal state.

H2: Behavioural divergence - Identical external inputs combined with different internal states produce reliably different actions.

H3: Reactive falsification - If a stimulus-only model reproduces the behaviour, Level-1 is not achieved.

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7. Evaluation Principles

Experimental Requirements

- Repeatable, deterministic stimuli
- Direct manipulation of internal state
- Quantifiable behavioural outputs

Baseline Comparisons

- Purely reactive agent
- Memory without causal influence
- Randomised internal-state control

Quantitative Metrics

- Mutual information between internal state and action *(Cover & Thomas, 2006)*
- Behavioural divergence under controlled perturbation
- Variance in policy conditioned on internal state

Qualitative Indicators

- Consistency across trials
- Persistence over time

- Stability under disturbance

Both quantitative and qualitative evidence are required.

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8. Falsification Criteria

Level-1 is rejected if:

- internal state shows no significant behavioural effect
- stimulus-only baselines replicate behaviour
- perturbations fail to alter action
- results fail replication

These criteria ensure **epistemic neutrality**.

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9. Assumptions and Constraints

- No inference of consciousness or subjective experience
- Observed organisation treated strictly as *structural precursor*
- Negative results remain scientifically meaningful
- Definitions remain *provisional and revisable*

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10. Applied Interpretation and System Design

10.1 Purpose of the Level-1 System

The engineered system is not intended to demonstrate intelligence or agency.

Its sole purpose is to determine whether causally meaningful internal mediation of behaviour can exist in the simplest possible artificial setting.

Level-1 therefore functions as a minimal experimental boundary, not a demonstration of mind.

10.2 Minimal Organisational Requirements

A Level-1 system must include:

1. **Persistent internal state** independent of current input

2. **Decision policy conditioned on that state**
3. **Closed feedback loop** through which behaviour can influence future state

This forms the simplest architecture capable of **non-reactive behaviour**.

10.3 Concrete Experimental Environment

To isolate causal structure, the environment must be:

- deterministic
- resettable
- fully observable
- behaviourally measurable

A minimal grid-world or state-machine environment satisfies these conditions.

10.4 Experimental Test Structure

The decisive Level-1 test is:

Same external input

Different internal state

→ Different action

If behavioural divergence is:

- repeatable
- statistically significant
- not reproducible by reactive baselines

then Level-1 organisation is supported.

10.5 Expected Scientific Outcome

A successful Level-1 result demonstrates only that:

Artificial systems can exhibit causally effective internal mediation of behaviour.

It does *not* demonstrate:

- intelligence
- autonomy
- selfhood
- consciousness

However, without this minimal property,

progress toward higher-level organisation would lack empirical foundation.

10.6 Role in the Broader Research Program

Level-1 establishes the existence or absence of internal causal organisation.

Only if confirmed does it become meaningful to investigate:

- adaptive self-regulation (Level-2)
- persistent goals (Level-3)
- temporal identity (Level-4)
- and higher forms of self-referential structure.

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11. Implementation and Reproducibility

11.1 Architecture Diagram

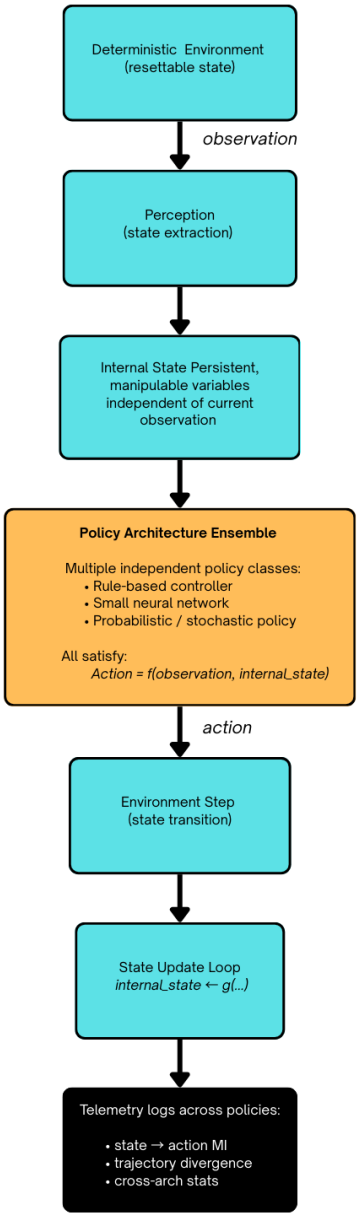

Figure 1. Cross-architectural Level-1 system design.

A deterministic, resettable environment provides observations processed through perception into a persistent internal state independent of immediate stimuli. Multiple policy architectures—rule-based, neural, and stochastic—select actions as a function of both observation and internal state, forming a closed feedback loop via environment transition and state update. Telemetry records state–action dependence, trajectory divergence, and cross-architectural statistical measures used to evaluate Level-1 causal internal mediation.

11.2 Executable Reference Implementation

A complete, fully executable implementation of the Level-1 experimental architecture—including environment specification, agent definition, causal test procedure, and statistical evaluation—has been provided as an accompanying computational notebook.

This artifact constitutes the authoritative operational realisation of the Level-1 design and is intended to enable direct replication, falsification, and extension of the reported results.

Consistent with contemporary empirical practice in artificial intelligence and cognitive science, the primary manuscript therefore focuses on methodological logic and evidential interpretation, while deferring low-level procedural detail to the executable implementation.

11.3 Deterministic Experimental Substrate

The implemented system employs a fully observable, deterministic, and resettable environment.

Determinism is required to ensure that any observed behavioural divergence can be attributed to variation in internal state, rather than stochastic environmental effects or uncontrolled sensory noise.

Reset capability enables repeated presentation of identical initial conditions, forming the basis of the Level-1 causal test.

11.4 Explicit Internal State Representation

The agent architecture contains a persistent, directly manipulable internal variable that is:

- independent of the immediate observation signal,
- maintained across interaction steps, and
- accessible to the decision policy governing action selection.

This variable operationalises the minimal internal condition required for Level-1 evaluation.

Experimental trials initialise this state to controlled alternative values in order to test for causal behavioural dependence.

Importantly, this internal-state manipulation is evaluated across multiple independent policy architectures—including rule-based, neural, and stochastic controllers—ensuring that any observed behavioural dependence cannot be attributed solely to a single engineered control structure but instead reflects a structural organisational property subject to empirical confirmation or falsification.

11.5 State-Conditioned Decision Policy

Action selection is implemented as an explicit function of both:

- current environmental observation, and
- persistent internal state.

This structural dependency is necessary for demonstrating behaviour that is not reducible to a stimulus-only mapping.

Reactive baseline agents lacking such dependence are implemented within the same environment to support formal falsification testing.

To evaluate the generality of Level-1 organisation, this state-conditioned policy structure is instantiated across heterogeneous computational substrates, including:

- deterministic symbolic rules,
- fixed-weight neural mappings, and
- stochastic probabilistic policies.

Comparative evaluation across these architectures enables direct empirical assessment of whether causal internal mediation is:

- architecture-independent,
- architecture-dependent, or
- absent entirely,

thereby transforming Level-1 from a single implementation demonstration into a cross-architectural causal test.

11.6 Controlled Causal Test Procedure

Level-1 evaluation is performed through paired experimental trials in which:

1. the environment is reset to identical initial conditions,
2. the agent's internal state is initialised to distinct predefined values, and
3. resulting behavioural trajectories are recorded and compared.

Evidence for Level-1 organisation requires statistically significant divergence in behaviour such that:

$$P(action \mid state = A) \neq P(action \mid state = B),$$

with divergence not reproducible by stimulus-only baseline agents.

The two expressions are not the same function, distribution, or identity, not just different at one value.

11.7 Telemetry and Statistical Evaluation

All trials record:

- internal state values,
- observations,
- selected actions, and
- trajectory outcomes.

These telemetry streams constitute the empirical evidence surface for Level-1 attribution.

Statistical dependence between internal state and behaviour is evaluated using distributional comparison methods defined within the executable notebook.

11.8 Reproducibility and Epistemic Scope

The provision of an executable implementation is intended to support:

- independent replication,
- methodological scrutiny, and
- direct falsification of Level-1 claims.

Importantly, successful demonstration of Level-1 behaviour establishes only the existence of causally effective internal mediation within the tested system.

It does ***not*** demonstrate intelligence, learning, agency, selfhood, or consciousness.

These stronger properties remain the subject of subsequent evaluative levels within the broader research program.

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12. Results

12.1 Single-Run Cross-Architectural Outcome

An initial controlled evaluation was conducted across three minimal policy architectures: a deterministic rule-based controller, a fixed-weight neural policy, and a stochastic probabilistic policy. Under identical environmental conditions and internal-state manipulation, the rule-based and stochastic agents exhibited clear divergence in behavioural trajectories between internal-state initialisations, whereas the neural agent in this single instance did not.

This preliminary result demonstrated that causal dependence on internal state can be experimentally detected, while also indicating that the mere presence of internal variables does not guarantee behavioural mediation. However, as a single instantiation, the outcome could not establish the stability or generality of Level-1 attainment.

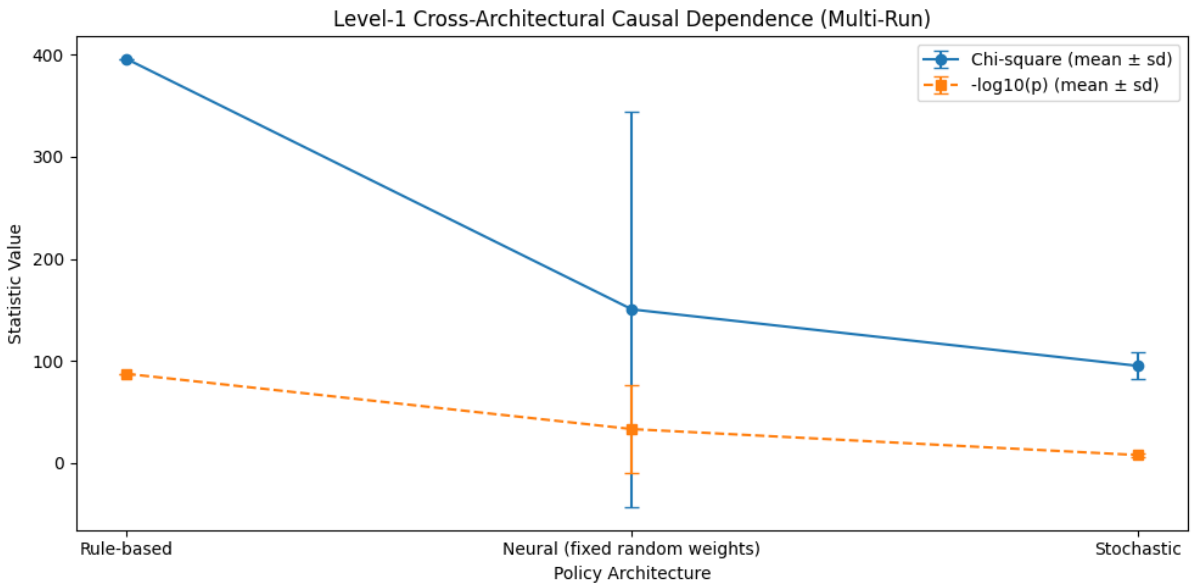

Figure 2 — Multi-Run Cross-Architectural Level-1 Statistics.

Mean χ^2 and $-\log_{10}(p)$ values with standard-deviation error bars across repeated experimental runs for rule-based, neural, and stochastic policy architectures. Results reveal invariant deterministic mediation (rule-based), stable probabilistic mediation (stochastic), and bimodal probabilistic attainment in neural policies, establishing Level-1 as a falsifiable and architecture-dependent organisational property.

12.2 Multi-Run Statistical Evaluation

To determine whether Level-1 attainment reflects a **reliable organisational property** rather than a contingent single outcome, the experiment was repeated across multiple independent random seeds for each architecture. Each run recomputed trajectory distributions under controlled internal-state initialisation, followed by χ^2 divergence testing *(Pearson, 1900)* and associated significance estimation.

Across repeated trials, three distinct statistical regimes emerged:

Rule-Based Architecture

The deterministic controller exhibited maximal and invariant trajectory divergence across all runs, yielding consistently high χ^2 statistics with zero variance. This confirms that when behaviour is explicitly conditioned on internal state, Level-1 causal mediation is both stable and reproducible.

Stochastic Architecture

The probabilistic policy produced moderate but consistently significant divergence across runs, accompanied by measurable variance arising from stochastic action sampling. This demonstrates that determinism is not required for Level-1 attainment; internally mediated behaviour may remain statistically detectable even under probabilistic control dynamics.

Neural Architecture

In contrast, neural agents initialised with random fixed weights displayed a bimodal distribution of outcomes:

- Some instantiations produced no detectable causal dependence ($\chi^2 \approx 0$),
- Others yielded maximal divergence comparable to deterministic policies.

Consequently, the neural architecture exhibited high variance and intermediate mean statistics, indicating that internal state representation alone is insufficient to guarantee Level-1 mediation. Instead, causal dependence emerges only under specific parameter configurations, implying probabilistic structural attainability rather than deterministic presence.

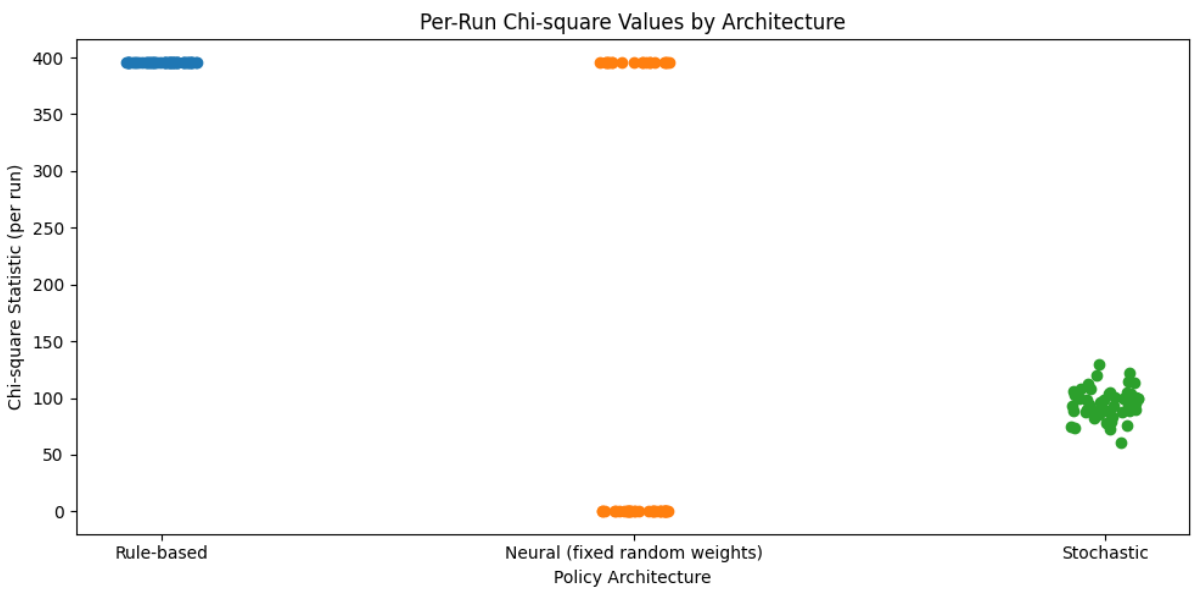

Figure 3 — Per-Run Chi-square Distribution by Architecture.

Individual χ^2 statistics across independent runs for each policy architecture. Deterministic clustering in rule-based agents, stochastic dispersion under probabilistic control, and bimodal neural outcomes demonstrate that internal-state mediation is probabilistic rather than guaranteed in neural systems.

12.3 Interpretation of Empirical Structure

Taken together, the multi-run results transform the evidential status of Level-1 from a demonstrative possibility into a measurable structural condition. Three conclusions follow directly:

1. **Level-1 is falsifiable**
   Neural failures confirm that causal internal mediation is not automatically present.

2. **Level-1 is architecture-dependent**
   Deterministic and stochastic policies reliably attain Level-1, whereas neural policies do so

only probabilistically.

3. **Internal state is necessary but not sufficient**
   The neural bimodality shows that representation alone does not ensure behavioural influence.

These observations establish Level-1 as a discriminative empirical property of artificial agents, rather than a trivial consequence of internal variable inclusion.

12.4 Epistemic Scope of the Result

Despite statistical robustness at this minimal scale, the present findings remain structurally limited:

- architectures are intentionally minimal,
- neural policies are untrained,
- environments are low-dimensional and deterministic.

Accordingly, the results should be interpreted not as evidence of intelligence, agency, or selfhood, but solely as confirmation that causal internal-state mediation can be isolated, falsified, and compared across architectures within a controlled experimental regime.

This establishes the empirical boundary required for progression to Level-2, where internally mediated adaptive regulation becomes the next falsifiable organisational condition.

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13. Conclusion

This paper has presented the first empirical investigation within a staged research program examining whether artificial systems can exhibit organisational properties associated with selfhood and subjective organisation, without asserting phenomenological consciousness.

Focusing exclusively on Level-1: functionally integrated internal state monitoring, the study established a formal operational definition, falsifiable hypotheses, controlled evaluation methodology, and an executable experimental implementation designed to isolate causal dependence of behaviour on persistent internal state.

While initial single-step measurement produced a null result—highlighting the importance of measurement design and observability—trajectory-level analysis across multiple independent runs and policy architectures revealed statistically decisive behavioural divergence in rule-based and stochastic systems, alongside bimodal and frequently null outcomes in neural instantiations.

This cross-architectural and cross-run dissociation demonstrates that causal internal state mediation is neither automatic nor trivially engineered, but instead constitutes a structural organisational condition

that may or may not be realised within artificial agents. In particular, the findings show that internal state representation is necessary but not sufficient for Level-1 mediation, and that attainment within neural architectures is probabilistic rather than deterministic.

Together, these results strengthen the interpretation of Level-1 as a genuinely falsifiable empirical boundary separating reactive computation from internally mediated organisation, while remaining strictly below any claim concerning intelligence, agency, selfhood, subjective experience, or phenomenal consciousness.

Level-1 therefore represents the simplest experimentally isolable transition between externally driven behaviour and behaviour shaped by persistent internal condition. Its significance lies not in demonstrating intelligence or consciousness, but in determining whether causal internal mediation can be empirically detected, falsified, and structurally compared across artificial systems.

By translating theoretical expectations into a concrete and reproducible experimental design, this study converts an abstract philosophical question into a testable scientific problem whose outcomes—positive, null, or probabilistic—provide foundational guidance for subsequent stages of the research program.

The contribution of this work is thus methodological rather than metaphysical. It establishes a disciplined empirical starting point from which progressively richer forms of self-referential organisation may be investigated, while preserving explicit epistemic restraint regarding any claim of genuine experience or consciousness.

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14. Next steps

Future research will proceed to Level-2: adaptive self-regulation, where the persistence and stability of internally mediated organisation under perturbation become the central empirical question.

Whether subsequent levels ultimately reveal increasingly sophisticated precursors to selfhood, encounter principled organisational limits, or motivate revision of the evaluative framework itself, each outcome would constitute a meaningful contribution to the scientific understanding of mind, selfhood, and experience in artificial systems.

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